

Analyzing emerging technology and skill trends

Nguyen Manh Ha
30/05/2026



OUTLINE



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EXECUTIVE SUMMARY



- **Staying Competitive in the IT Industry**
- **Study Overview:**
 - ❖ Analyzed current and future technology trends using Stack Overflow developer survey data.
 - ❖ Explored programming languages, databases, cloud platforms, web frameworks, and developer demographics.
- ☐ **Data Sources:**
 - ❖ Stack Overflow Developer Survey dataset.
- ☐ **Data Analysis Process:**
 - ❖ Data Collection → Data Cleaning → Exploratory Analysis → Dashboard Visualization.
- ☐ **Key Findings:**
 - ❖ Python is the most widely used and most desired programming language among developers.
 - ❖ PostgreSQL leads both current database adoption and future demand.
 - ❖ AWS remains the most preferred cloud platform, while React dominates web development.
- ☐ **Developer Demographics:**
 - ❖ Most respondents are 25–34 years old and hold Bachelor's or Master's degrees.
 - ❖ The United States contributes the largest share of survey respondents.



INTRODUCTION



- ❖ This dashboard analyzes current technology adoption and future skill demand using developer survey data, with a focus on programming languages, databases, cloud platforms, web frameworks, and workforce demographics.
- ❖ The objective is to identify the technologies that are most relevant in today's market and those expected to drive future industry growth.
- ❖ Key questions addressed in this analysis include:
 - Which programming languages are most widely used and most desired for the future?
 - Which databases, cloud platforms, and web frameworks show the strongest demand?
 - What demographic characteristics define the modern technology workforce?
- ❖ The findings provide valuable insights for recruiters, hiring managers, IT professionals, students, and career changers seeking to align their skills with current and emerging industry trends.
- ❖ By highlighting high-demand technologies and workforce patterns, this study supports data-driven decisions in recruitment, workforce planning, and professional development.



METHODOLOGY



► Data Collection:

- Collected and prepared Stack Overflow Developer Survey data covering technology usage, future preferences, and demographic information.

► Data Cleaning & Preparation:

- Processed missing values, standardized categorical variables, and transformed multi-response fields for analysis.

► Exploratory Data Analysis (EDA):

- Examined technology adoption patterns, future skill demand, and respondent demographics to identify key trends.

► Data Visualization:

- Created charts and comparative visualizations to highlight technology popularity, growth opportunities, and workforce characteristics.

► Dashboard Development:

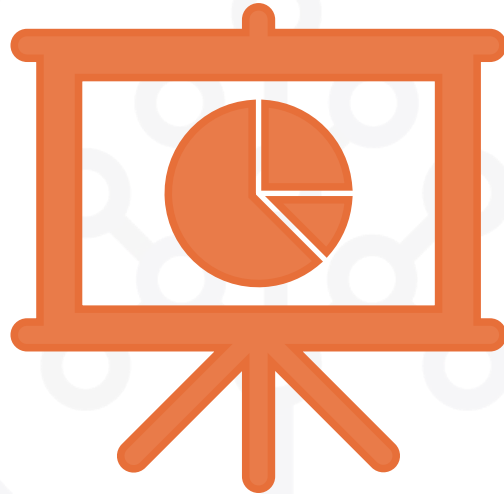
- Built an interactive dashboard in Google Looker Studio to present insights on programming languages, databases, cloud platforms, web frameworks, and developer demographics.

► Insight Generation:

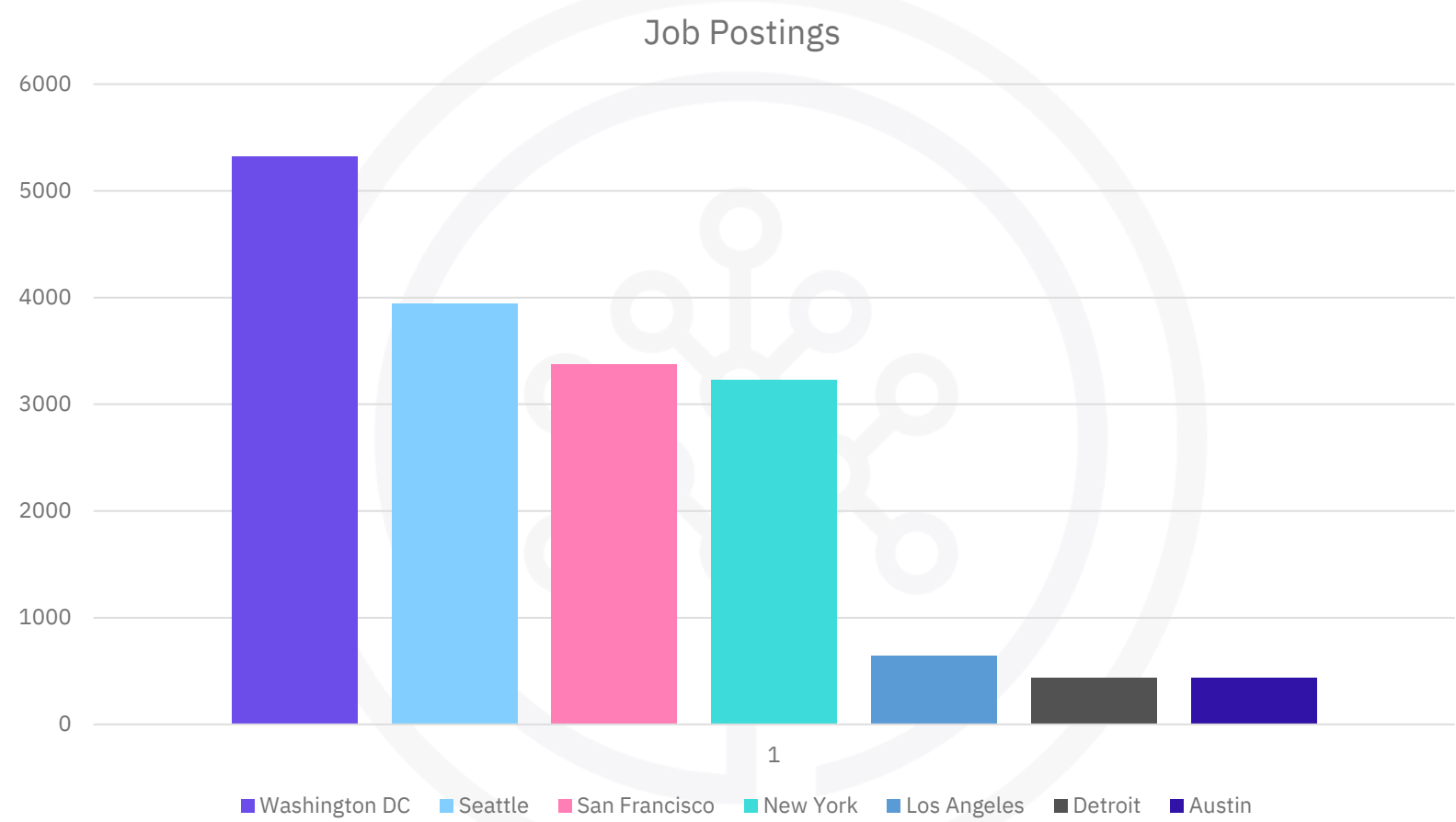
- Interpreted findings to identify high-demand technologies and emerging skills that support recruitment, career planning, and workforce development.



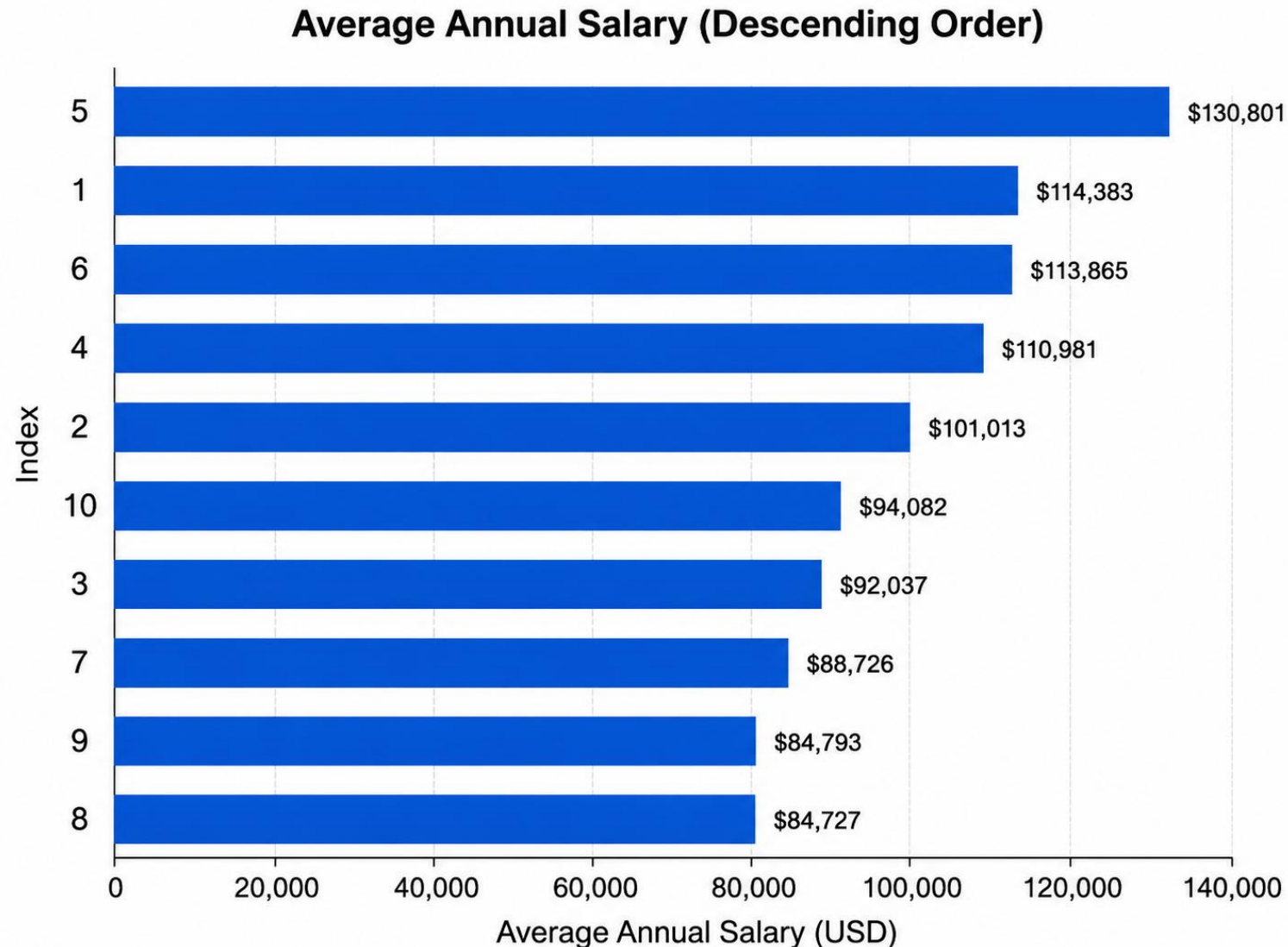
Results



JOB POSTINGS



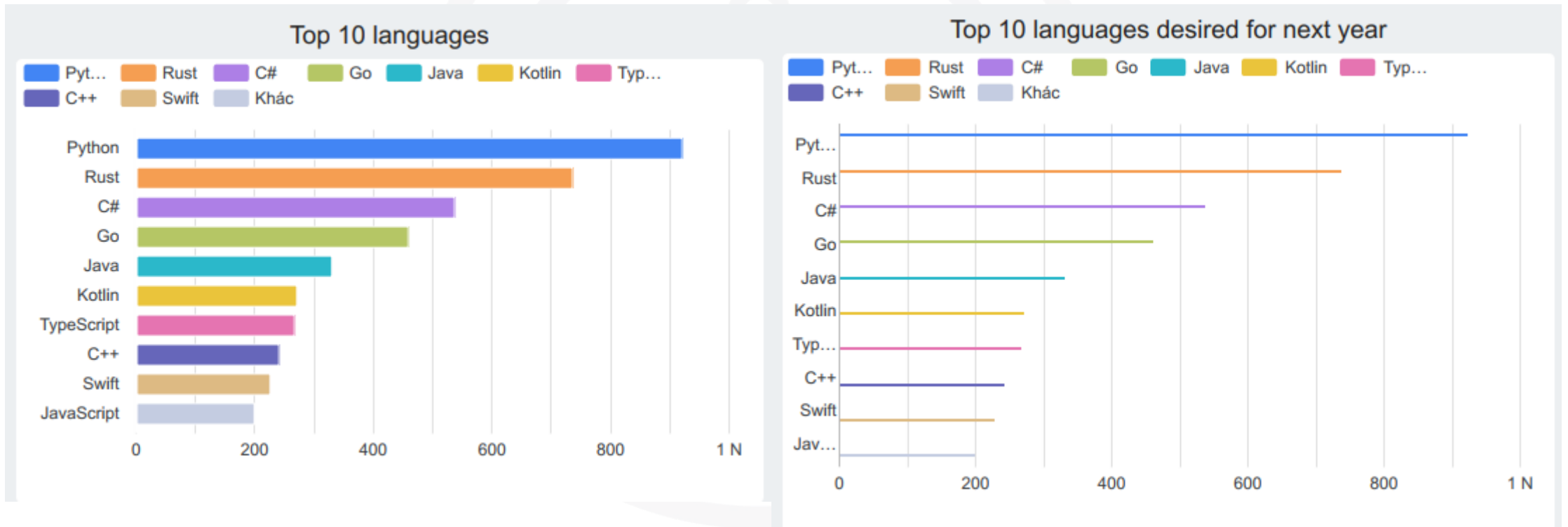
POPULAR LANGUAGES



PROGRAMMING LANGUAGE TRENDS

Current Year

Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Python remains the strongest programming language in both current usage and future demand.
- Rust shows very strong future interest.
- JavaScript drops in future preference compared with current usage.

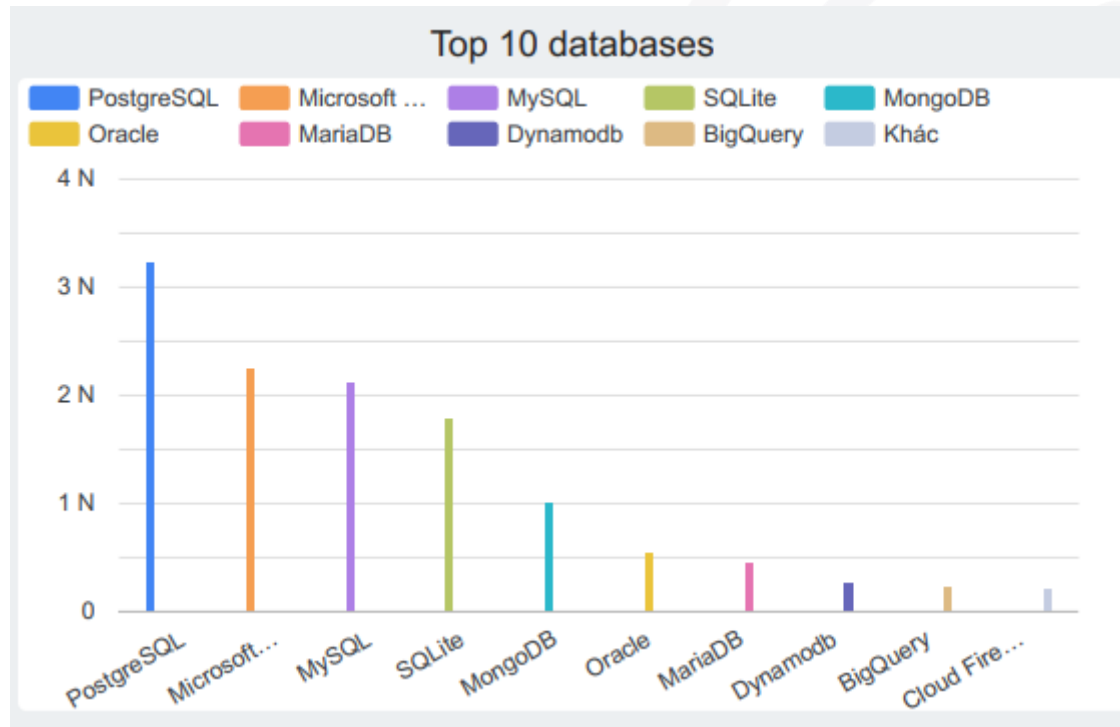
Implications

- Python should remain a core skill for data, AI, and software roles.
- Rust may become a valuable differentiator.
- Developers should balance established and emerging skills.

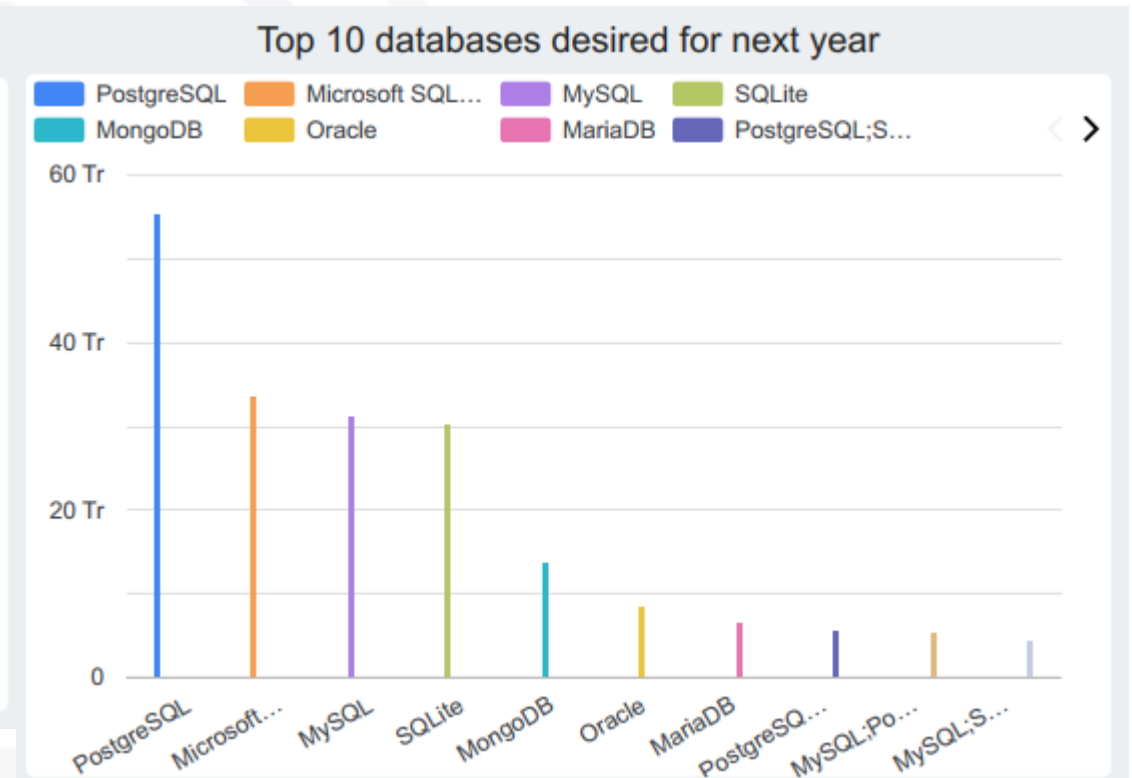


DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the leading database in both current usage and future demand.
- Microsoft SQL Server, MySQL, and SQLite remain major database choices.
- MongoDB shows stronger future interest compared with smaller databases.

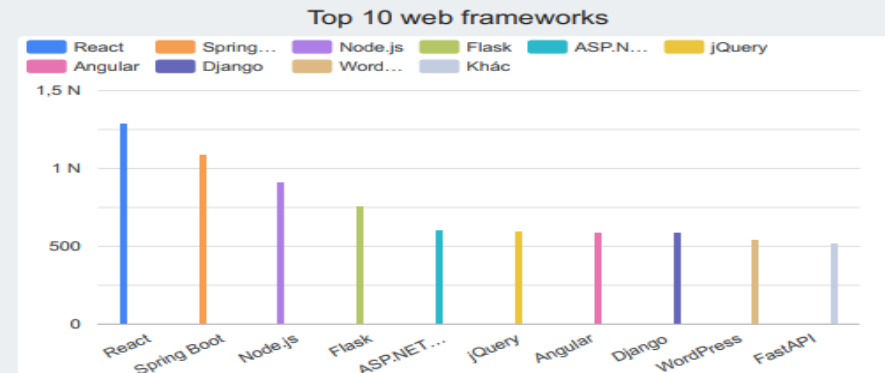
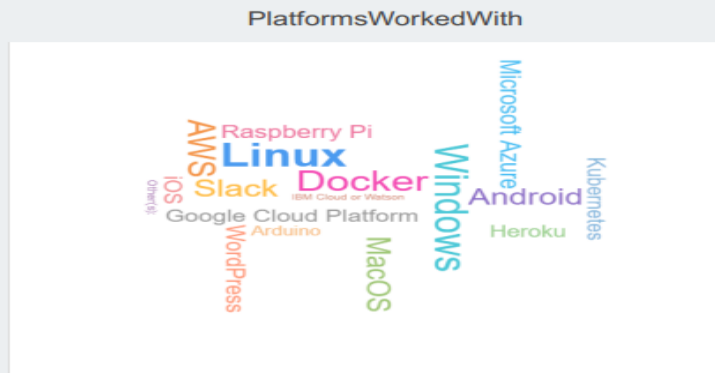
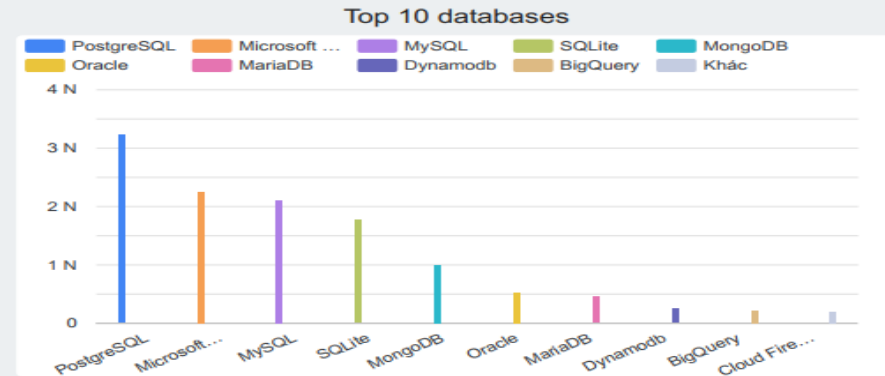
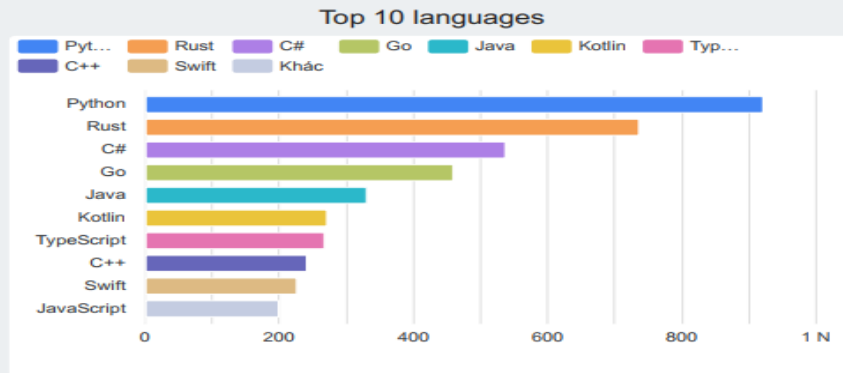
Implications

- SQL-based databases should remain a learning priority.
- PostgreSQL is especially valuable for career development.
- Learners should combine relational and NoSQL knowledge.

DASHBOARD



DASHBOARD TAB 1: Current Technology Usage



DASHBOARD TAB 1: Current Technology Usage

- **Dashboard Insight**

Python, PostgreSQL, React, and cloud-related platforms are the strongest technologies in current usage.

The dashboard shows high demand for skills across programming, databases, platforms, and web frameworks.

This indicates that employers value candidates with a broad, practical technology stack.

- **Summary of Future Trends**

Future demand is expected to grow around Python, SQL databases, cloud platforms, and modern web frameworks.

PostgreSQL, React, and cloud tools are likely to remain important for data, backend, and full-stack roles.

Candidates should focus on building job-ready skills across Python, SQL, cloud, and web development tools.



DASHBOARD TAB 2: Future Technology Trends



DASHBOARD TAB 2: Future Technology Trends

- **Dashboard Insight**

The dashboard shows strong future interest in Python, PostgreSQL, AWS, and React across programming, databases, platforms, and web frameworks.

Cloud platforms, especially AWS and Microsoft Azure, stand out as key technologies for future development work.

Overall, the data suggests that future-ready candidates need skills across coding, databases, cloud, and web technologies.

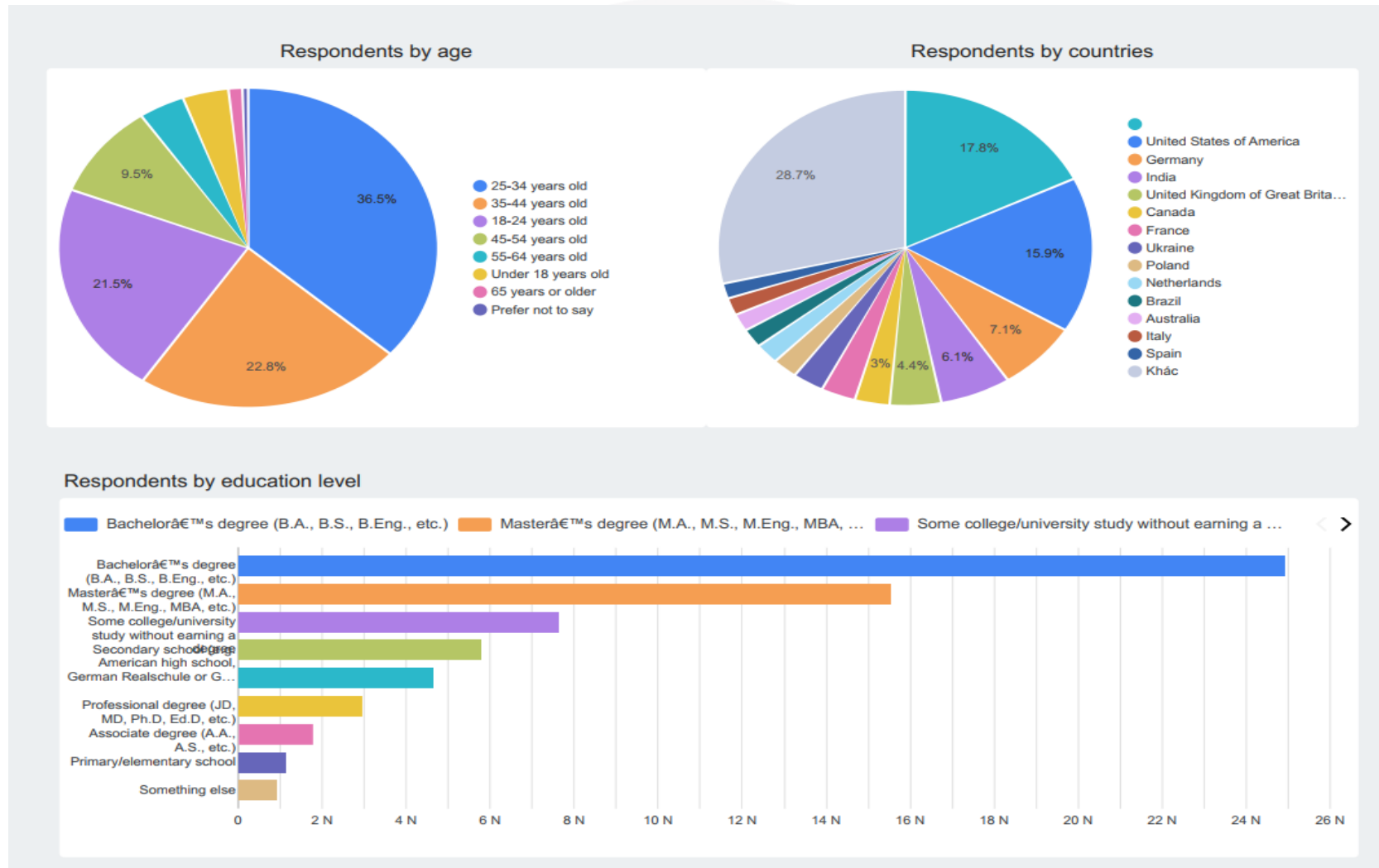
- **Summary of Future Trends**

Future technology demand is likely to grow around Python, SQL databases, cloud platforms, and modern web frameworks.

PostgreSQL, AWS, React, and Spring Boot appear especially important for backend, data, and full-stack roles.

Candidates who combine Python, SQL, cloud, and web framework skills will be better positioned for recruiter expectations.

DASHBOARD TAB 3: Demographics



DASHBOARD TAB 3: Demographics

- **Dashboard Insight**

The respondent base is concentrated in younger age groups, especially 25–34 years old, followed by 35–44 years old.

Most respondents come from the United States, with strong representation from Germany, India, the United Kingdom, and Canada.

Education levels are highly professional, with Bachelor's and Master's degrees making up the largest respondent groups.

- **Summary of Future Trends**

The tech workforce is likely to remain driven by young and early-to-mid career professionals.

Global participation suggests that technology skills are becoming increasingly important across both developed and emerging markets.

Recruiters should expect strong talent pools from candidates with formal higher education and internationally diverse backgrounds.

DISCUSSION

- **Overall insights:**

The dashboards show strong demand for a full-stack technology profile, with Python, PostgreSQL, AWS, and React standing out across current usage and future technology trends.

Future demand is shifting toward skills that support data analytics, cloud computing, backend development, and modern web applications.

The respondent base is mainly young, globally distributed, and highly educated, suggesting that competition in tech roles will remain strong and skill-based.

- **Overall implications:**

Candidates should prioritize a balanced skill set in Python, SQL/PostgreSQL, cloud platforms, and web frameworks.

For recruiter readiness, the strongest profile is not only one tool, but the ability to connect programming, databases, cloud, and analytics into real business solutions.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Python, PostgreSQL, AWS, and React are the strongest technologies across the dashboards.
- Future technology demand is moving toward cloud-based and full-stack skills.
- The respondent profile is young, global, and highly educated, with strong representation from the 25–34 age group and many respondents holding Bachelor's or Master's degrees.

Implications

- Candidates should build a balanced tech stack, combining Python, SQL/PostgreSQL, cloud platforms, and modern web frameworks.
- Cloud and database skills are becoming essential, not optional, for data analyst, backend, and full-stack career paths.
- Recruiters are likely to value practical, job-ready portfolios, especially projects that connect programming, databases, dashboards, and business insights.



CONCLUSION



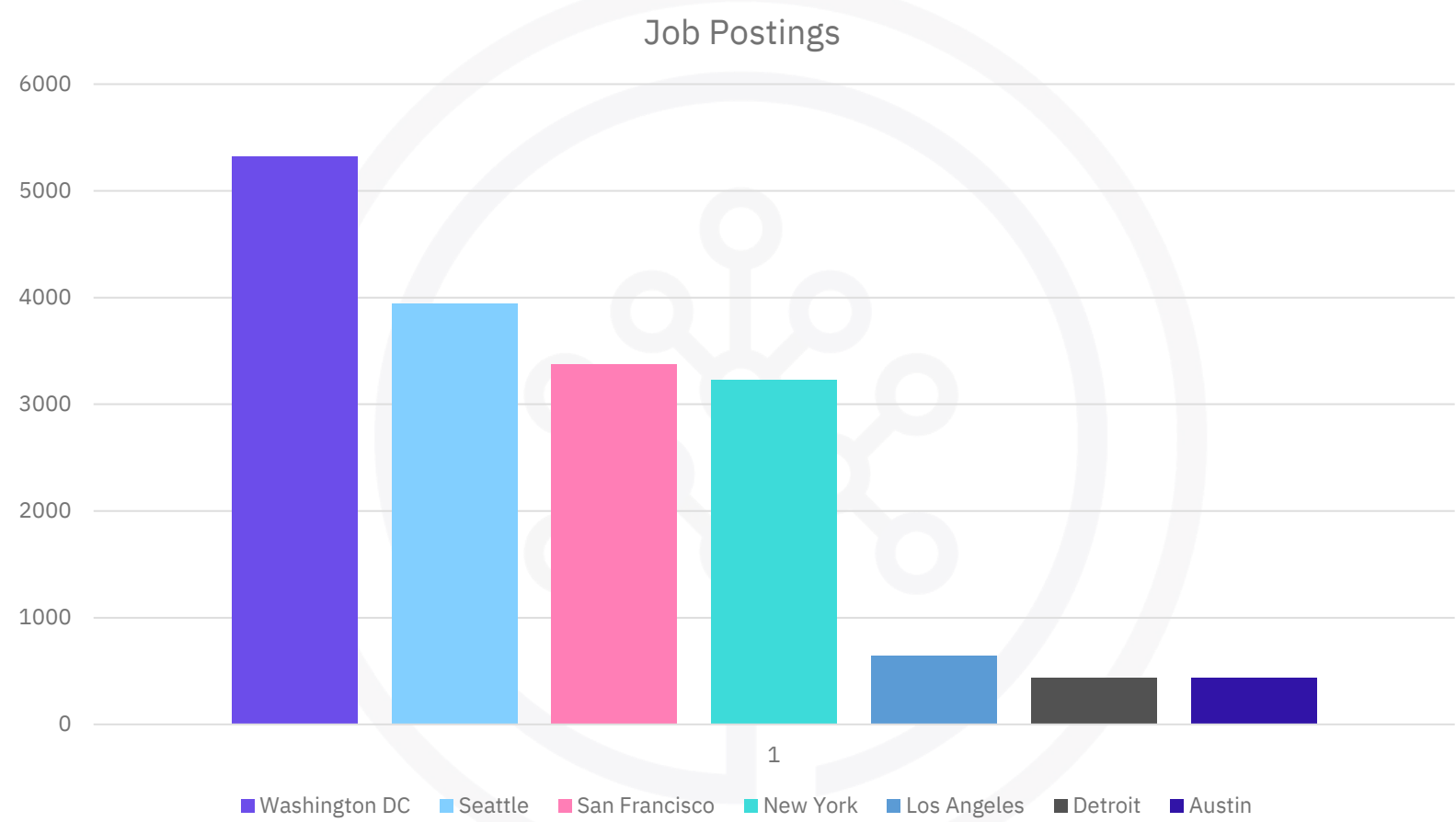
- Technology trends show that developers must continuously upgrade their skills to stay relevant in a changing market.
- Python, PostgreSQL, AWS, and React stand out as key technologies for future-ready tech roles.
- Cloud computing, databases, and web frameworks remain highly valuable areas for both current usage and future demand.
- A Bachelor's degree is common among respondents, but practical skills and a strong portfolio are essential for recruiter readiness.



APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

